⑤ Extracts the instance's ID from the CloudTrail event

⑥ Adds a tag to the EC2 instance using the key owner and the user's name as value

After implementing your function in Python, the next step is to deploy the Lambda function with all its dependencies.

6.3.3 Setting up a Lambda function with the Serverless Application Model (SAM)

You have probably noticed that we are huge fans of automating infrastructures with CloudFormation. Using the Management Console is a perfect way to take the first step when learning about a new service on AWS. But leveling up from manually clicking through a web interface to fully automating the deployment of your infrastructure should be your second step.

AWS released the *Serverless Application Model* (SAM) in 2016. SAM provides a framework for serverless applications, extending plain CloudFormation templates to make it easier to deploy Lambda functions. The next listing shows how to define a Lambda function using SAM and a CloudFormation template.

Listing 6.5 Defining a Lambda function with SAM within a CloudFormation template

```
---
AWSTemplateFormatVersion: '2010-09-09'                          ①
Transform: AWS::Serverless-2016-10-31                           ②
Description: Adding an owner tag to EC2 instances automatically
Resources:
  # [...]
  EC2OwnerTagFunction:
    Type: AWS::Serverless::Function                             ③
    Properties:
      Handler: lambda_function.lambda_handler                   ④
```